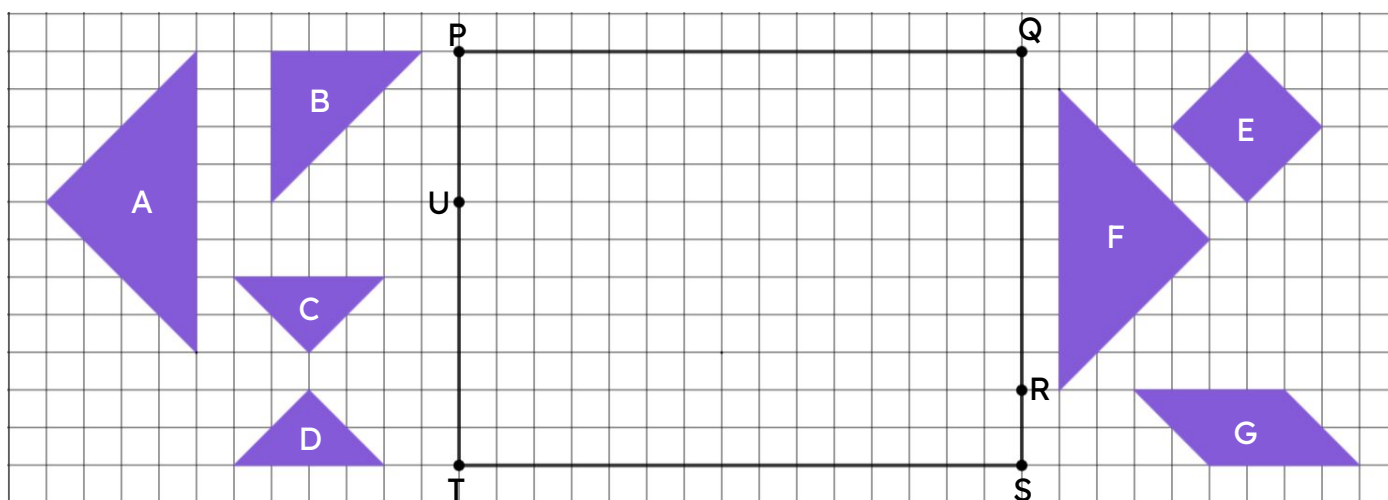




Checking understanding of congruence

Task A: Transformations with congruence

- 1) a) Reflect A over PT. b) Rotate B 180° around U. c) Translate C $\begin{pmatrix} 6 \\ -3 \end{pmatrix}$
d) Reflect D over PT. e) Translate E $\begin{pmatrix} -15 \\ -7 \end{pmatrix}$
f) Rotate F 90° anticlockwise around R. g) Reflect G over QS



Task B: Identifying missing information on congruent shapes

1) The five triangles are congruent.

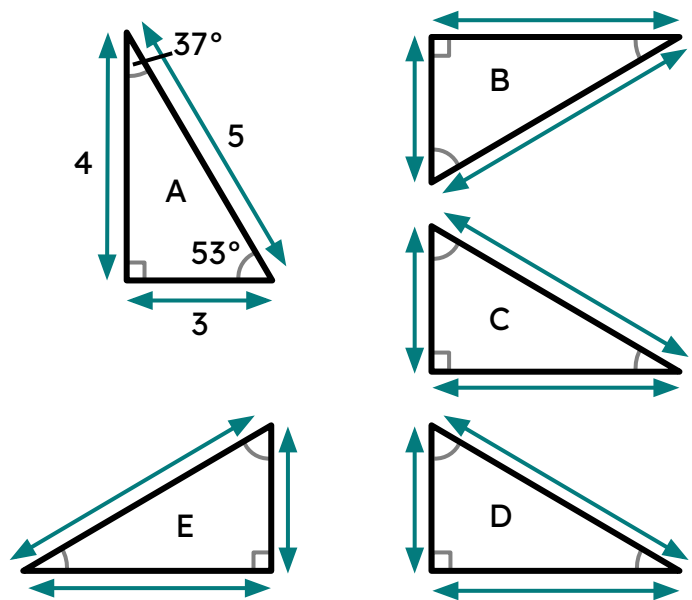
a) What transformation maps A onto B?

b) What transformation maps B onto C?

c) What transformation maps C onto D?

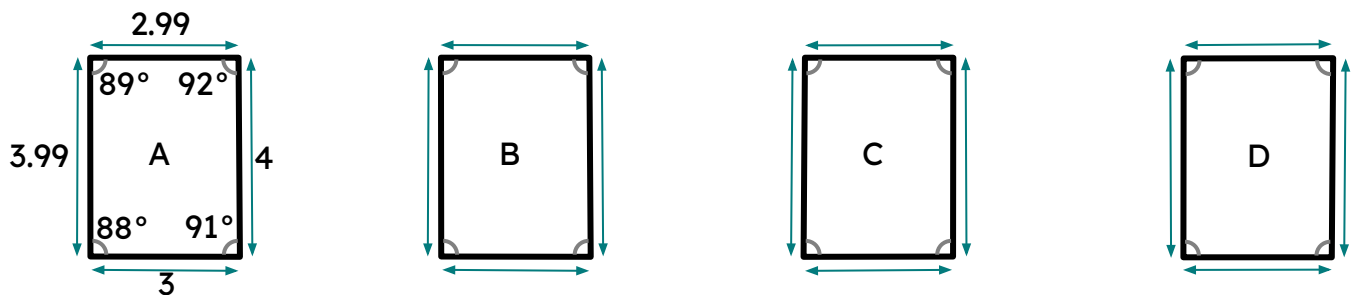
d) What transformation maps D onto E?

e) What transformation maps E onto A?



f) Label the missing lengths and angles on triangles B to E.

2) The four quadrilaterals are congruent. The transformations which map each shape onto the next shape are stated below.



A $\rightarrow$ B: translation

B $\rightarrow$ C: reflection

C $\rightarrow$ D: rotation of 180°

a) Label the missing lengths and angles on shapes B to D.

b) Which transformation could be applied to shape D to create an image that is in the same orientation as shape A?